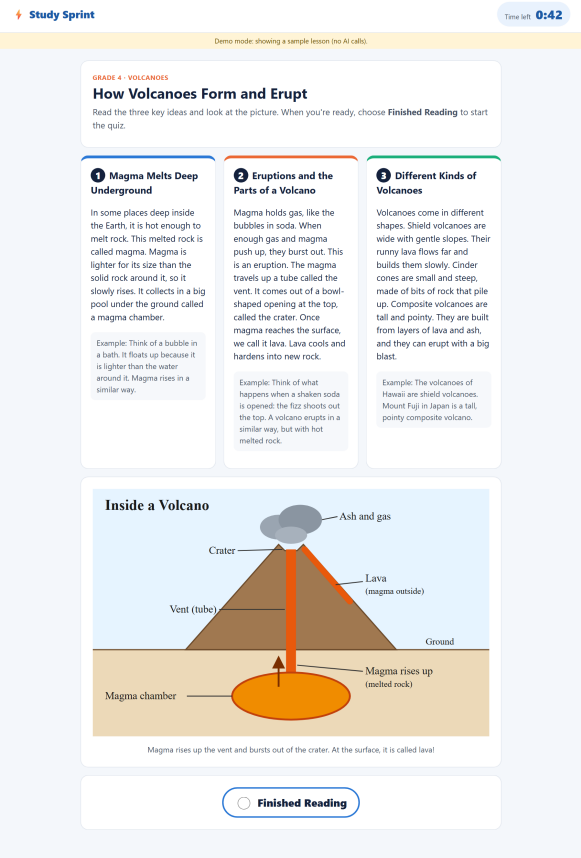
Study Sprint

How the app works

A multi-agent study app: pick a school subject and grade, read three key ideas with a diagram, take a 5-question quiz, and do a quick movement break whenever the 1-minute timer runs out.



The reading screen for a lesson built by the agents (volcanoes, Grade 4; explanations lightly edited).

What it does

Study Sprint turns any school subject into a short lesson at the right level for the learner, from Kindergarten to College. Each lesson has **three important concepts**, a **labeled diagram** and a **5-question quiz**. Five Claude agents write the lesson; code checks everything they write before a learner sees it.

Movement breaks. The reading screen and the quiz each run a 60-second timer. When it reaches zero, the app picks one exercise at random (10 jumping jacks, 10 high knees (running in place), 10 push-ups). The learner clicks *I did it!* and the timer starts again. The results screen lists every exercise completed.

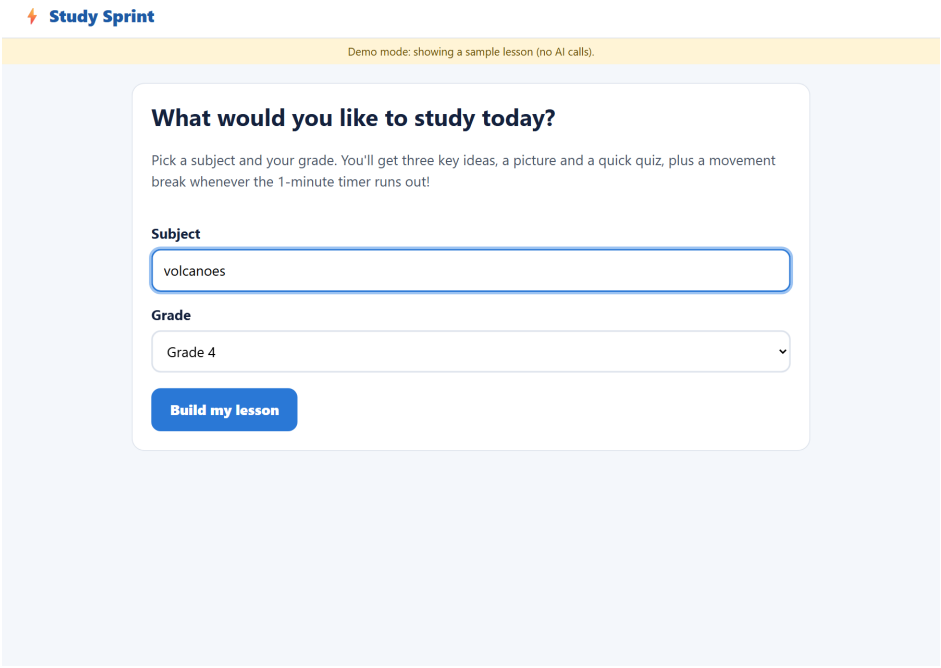
Built to be safe for children. Subjects are checked before any agent runs. Every sentence the agents write is checked for links, requests for personal information and unkind words. Diagrams are sanitized, and answer keys never reach the browser until the quiz is graded. No names or accounts are collected.

The session, step by step

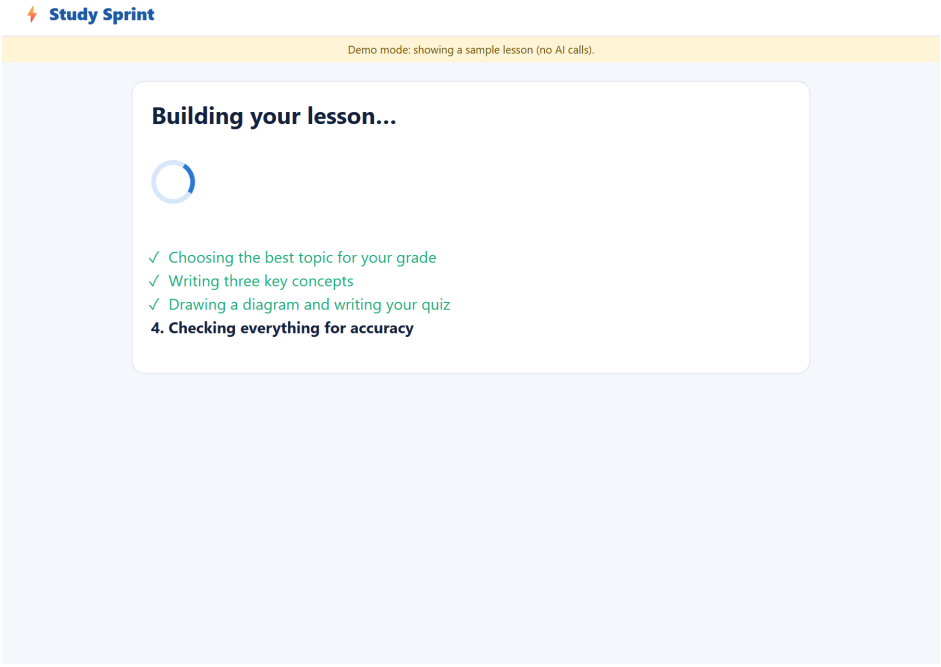
Step	What happens	Who does it
1. Choose	Type a subject, pick a grade, press *Build my lesson*.	Learner
2. Build	Topic gate, then five agents write and check the lesson (about a minute).	Code + agents
3. Read	Three concepts and a diagram, with a 60-second timer.	Learner
4. Move	Timer at zero: a random exercise, then the timer restarts.	Code picks; learner moves
5. Quiz	Select *Finished Reading*; answer 5 questions (same timer and breaks).	Learner
6. Celebrate	Score, percentage, a big "Great work!", the exercises done and an answer review. *Start Over* asks for a new subject and grade.	Code grades

Walkthrough

These screenshots were taken automatically (`run.ps1 guide`) in **demo mode**, which replays a lesson saved from a live agent build. That's why each screen shows a demo banner: the guide can be regenerated at any time without API calls, and the content is what the agents actually produced.



1. The start screen asks what to study and the learner's grade (Kindergarten to College).



2. While the agents work (about a minute live), the progress screen lists each stage.

Study Sprint

Time left 0:42

Demo mode: showing a sample lesson (no AI calls).

GRADE 4 · VOLCANOES

How Volcanoes Form and Erupt

Read the three key ideas and look at the picture. When you're ready, choose **Finished Reading** to start the quiz.

1 Magma Melts Deep Underground

In some places deep inside the Earth, it is hot enough to melt rock. This melted rock is called magma. Magma is lighter for its size than the solid rock around it, so it slowly rises. It collects in a big pool under the ground called a magma chamber.

Example: Think of a bubble in a bath. It floats up because it is lighter than the water around it. Magma rises in a similar way.

2 Eruptions and the Parts of a Volcano

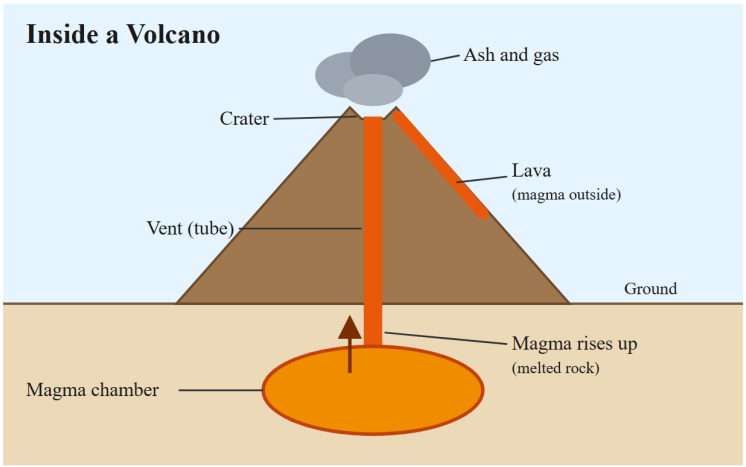
Magma holds gas, like the bubbles in soda. When enough gas and magma push up, they burst out. This is an eruption. The magma travels up a tube called the vent. It comes out of a bowl-shaped opening at the top, called the crater. Once magma reaches the surface, we call it lava. Lava cools and hardens into new rock.

Example: Think of what happens when a shaken soda is opened: the fizz shoots out the top. A volcano erupts in a similar way, but with hot melted rock.

3 Different Kinds of Volcanoes

Volcanoes come in different shapes. Shield volcanoes are wide with gentle slopes. Their runny lava flows far and builds them slowly. Cinder cones are small and steep, made of bits of rock that pile up. Composite volcanoes are tall and pointy. They are built from layers of lava and ash, and they can erupt with a big blast.

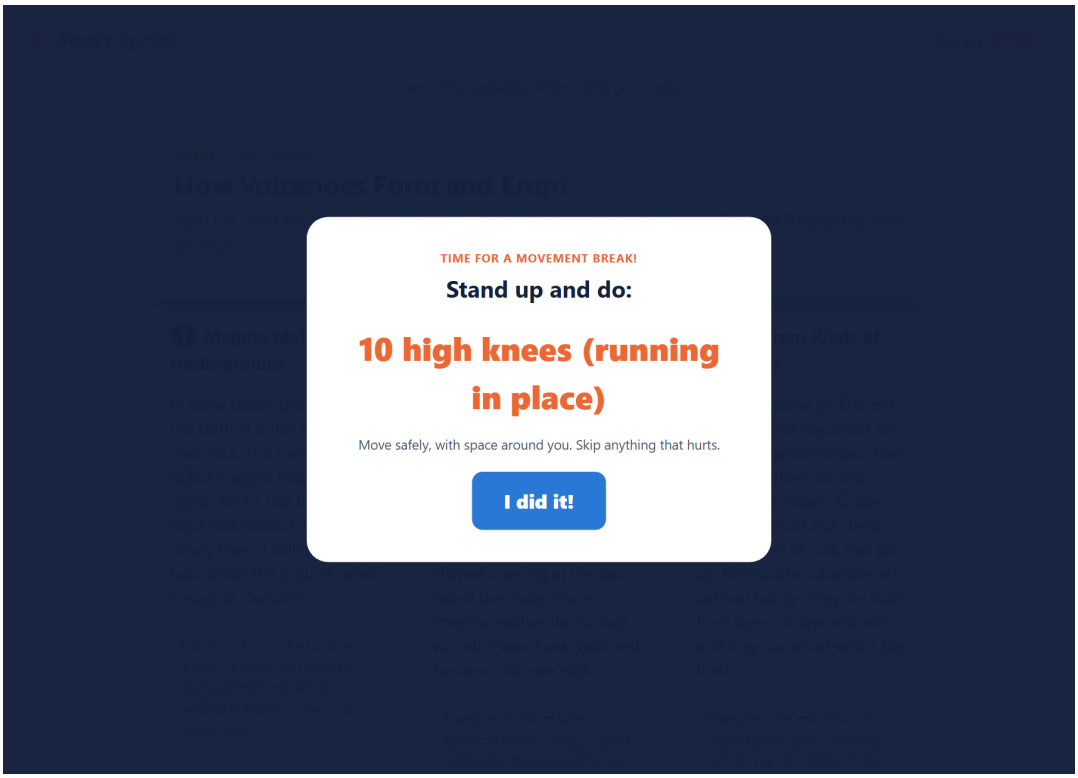
Example: The volcanoes of Hawaii are shield volcanoes. Mount Fuji in Japan is a tall, pointy composite volcano.



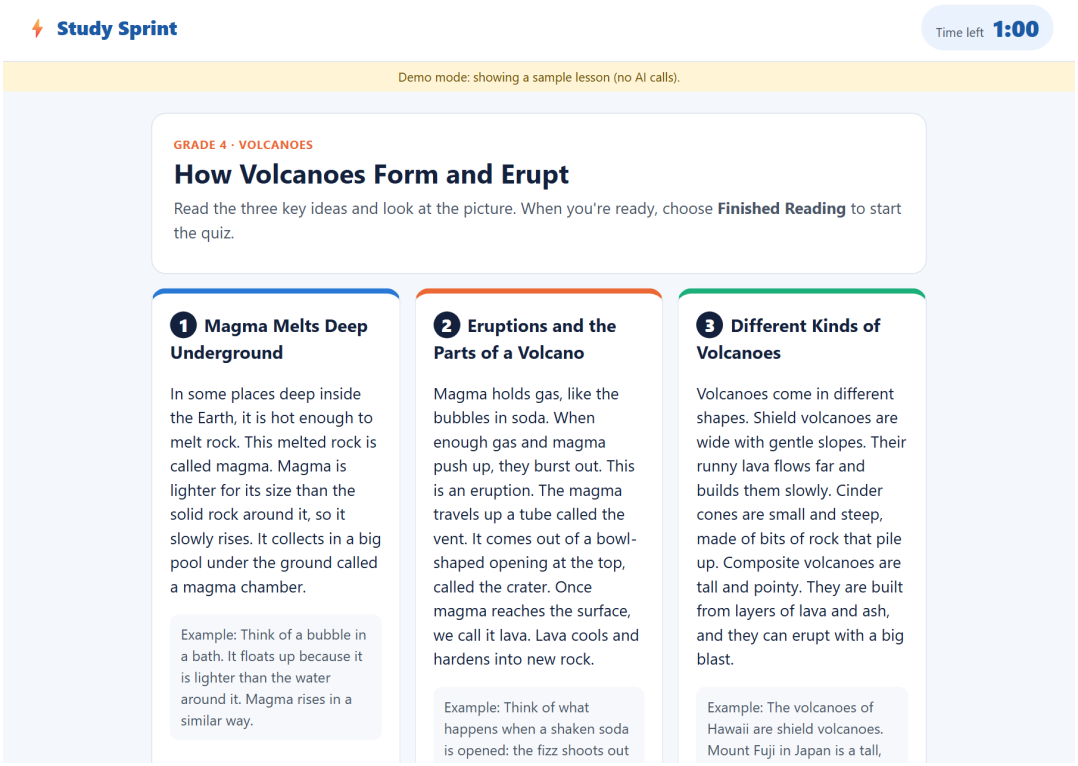
Magma rises up the vent and bursts out of the crater. At the surface, it is called lava!

Finished Reading

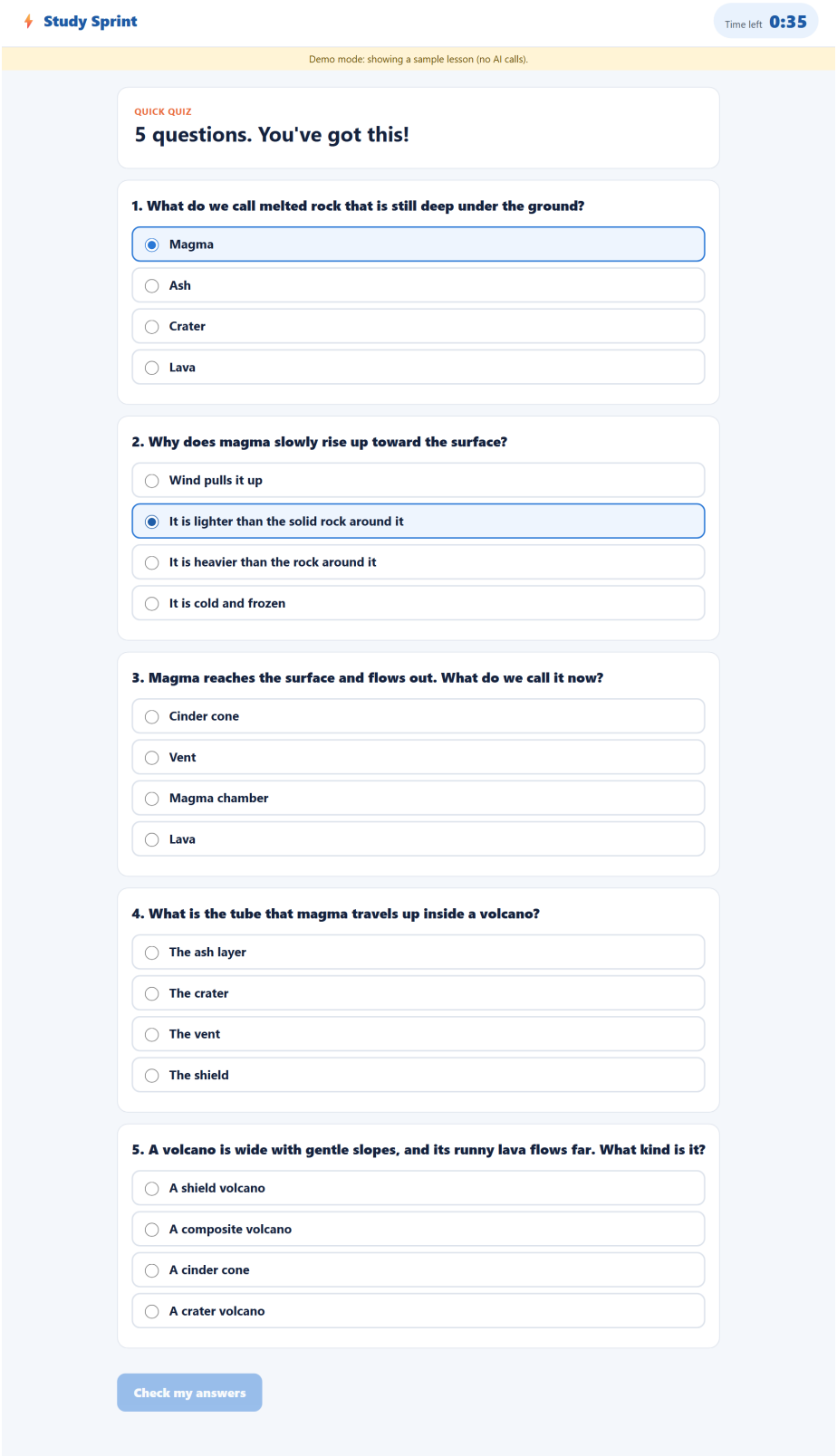
3. The lesson: three key concepts, each with an example, and a diagram drawn by the Diagram Agent. The timer (top right) counts down from 1:00. Selecting **Finished Reading** opens the quiz.



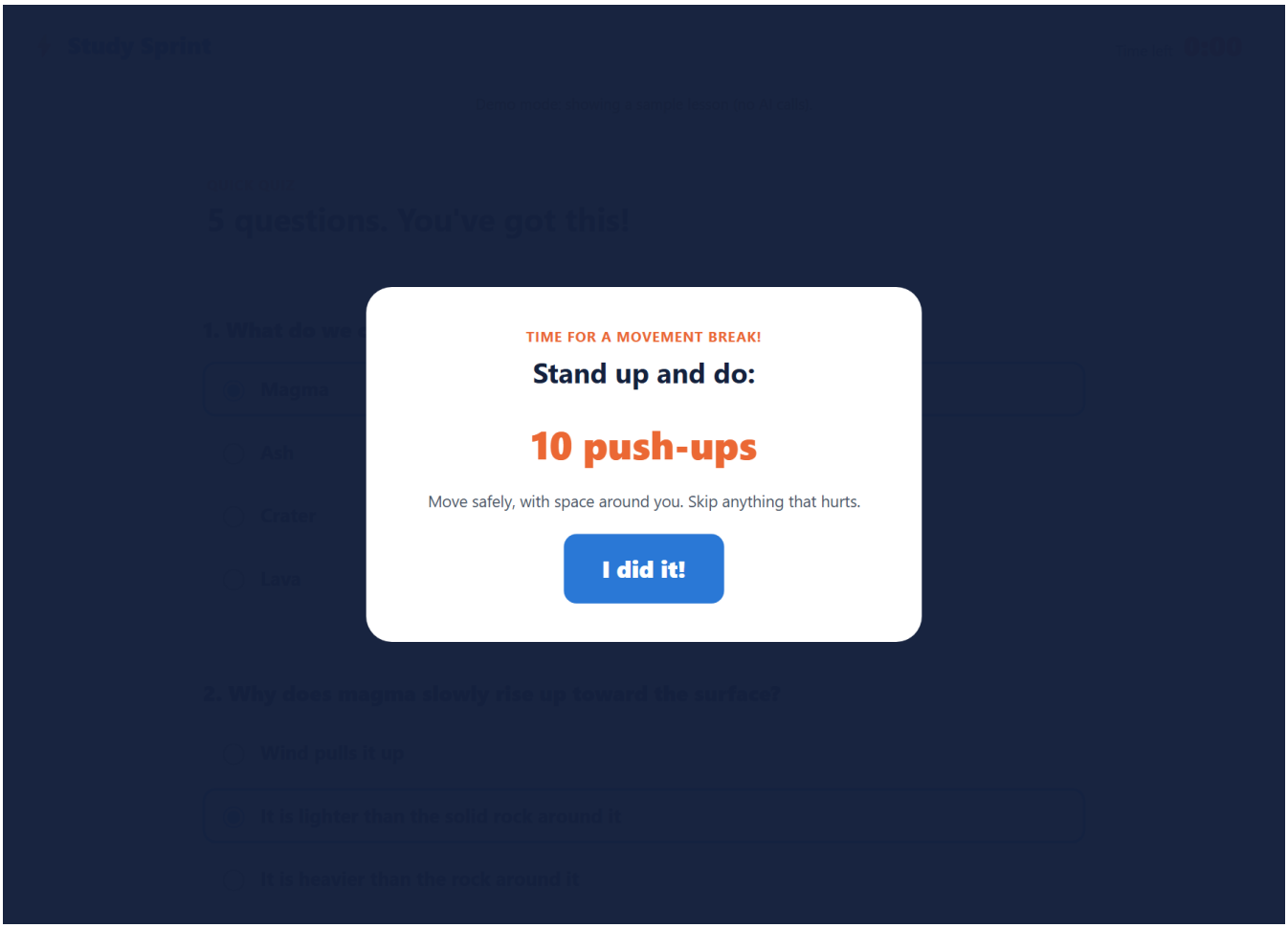
4. When the minute is up, the lesson is covered and a random exercise appears.



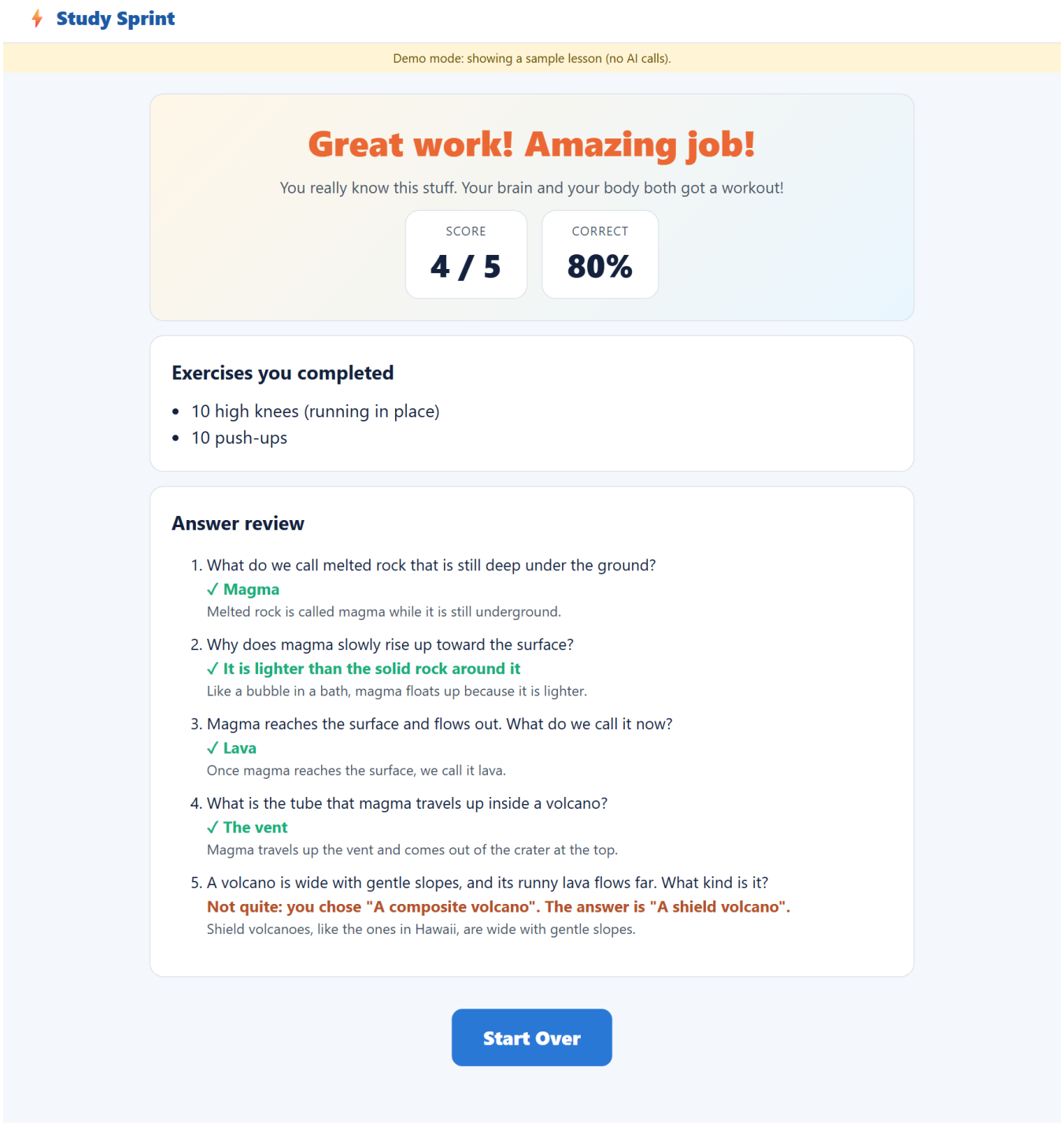
After *I did it!*, the lesson returns and the timer starts again at 1:00.



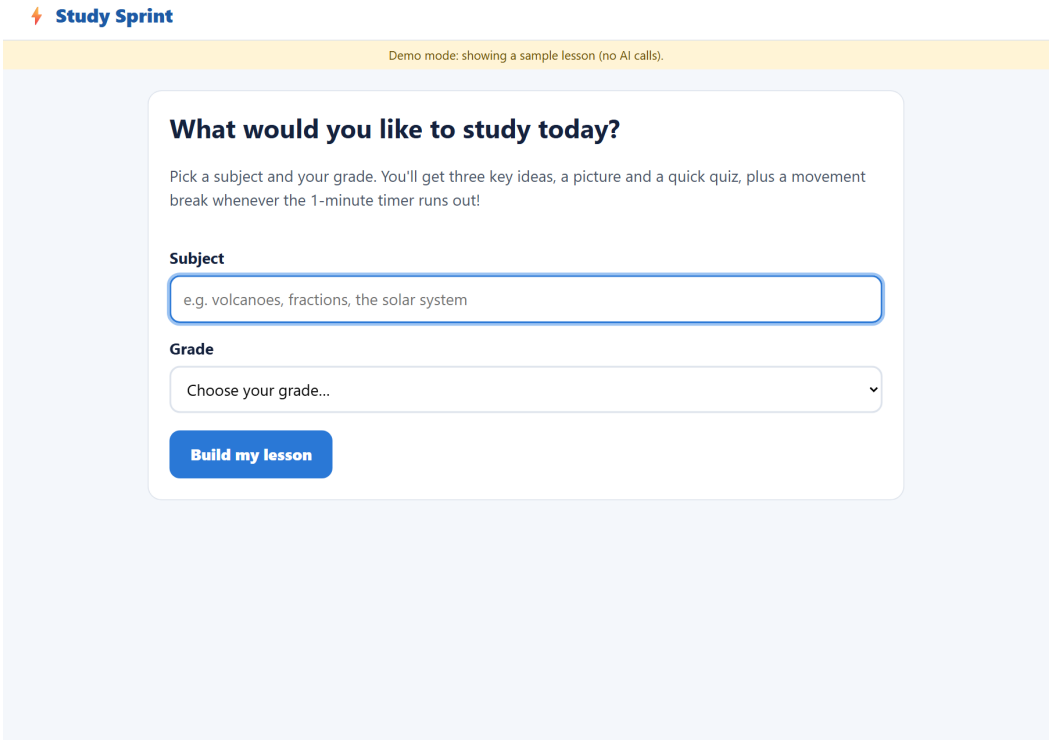
5. The quiz: five multiple-choice questions on the concepts, then *Check my answers*. The answer key stays on the server until grading.



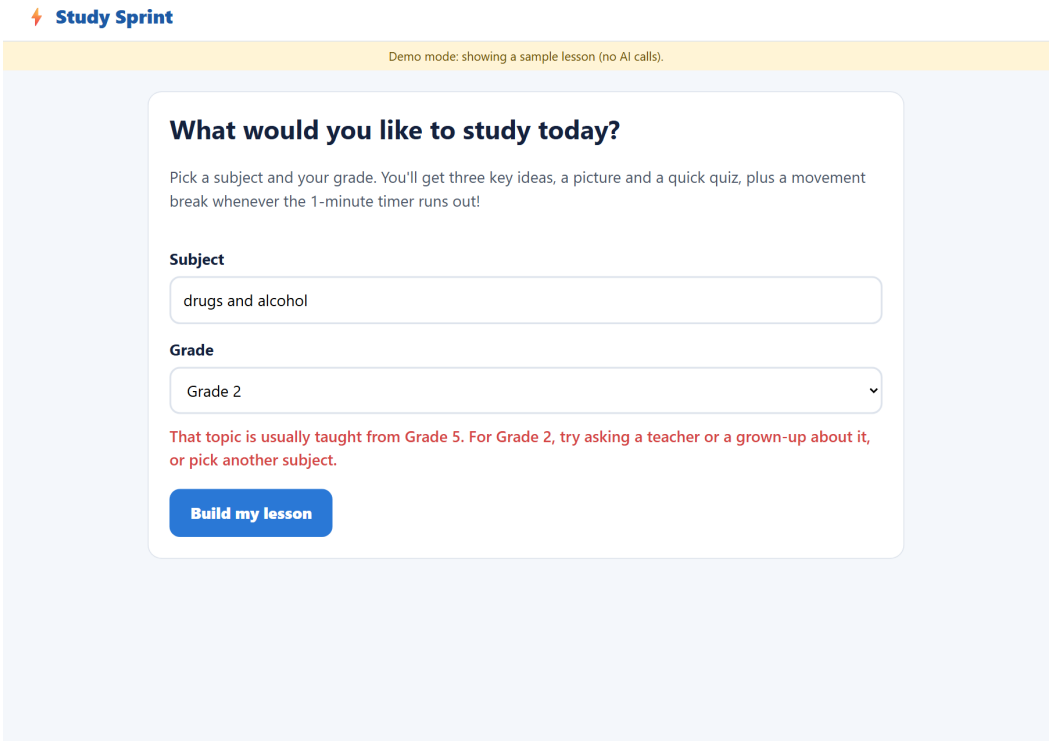
6. A minute passing during the quiz brings another movement break before continuing. Each break's exercise is picked at random, so it often differs from the last one.



7. Results: score, percentage, an enthusiastic message, every exercise completed and a gentle answer review.

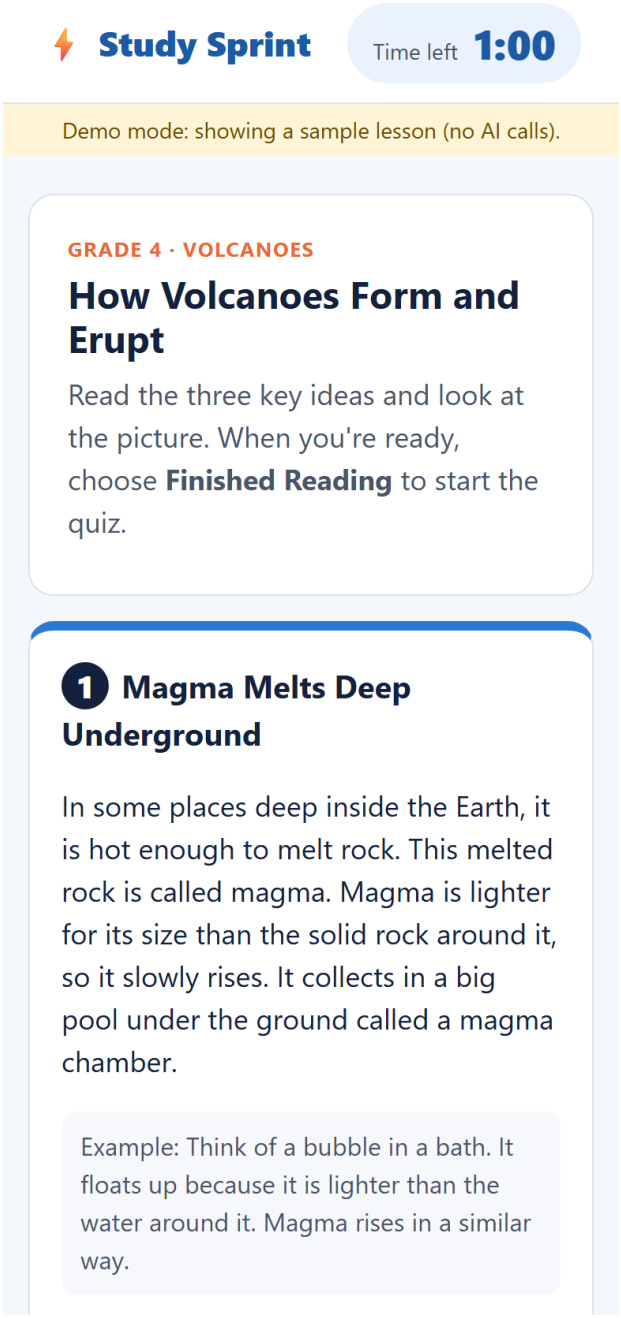


8. *Start Over* returns to an empty form, ready for a new subject and grade.



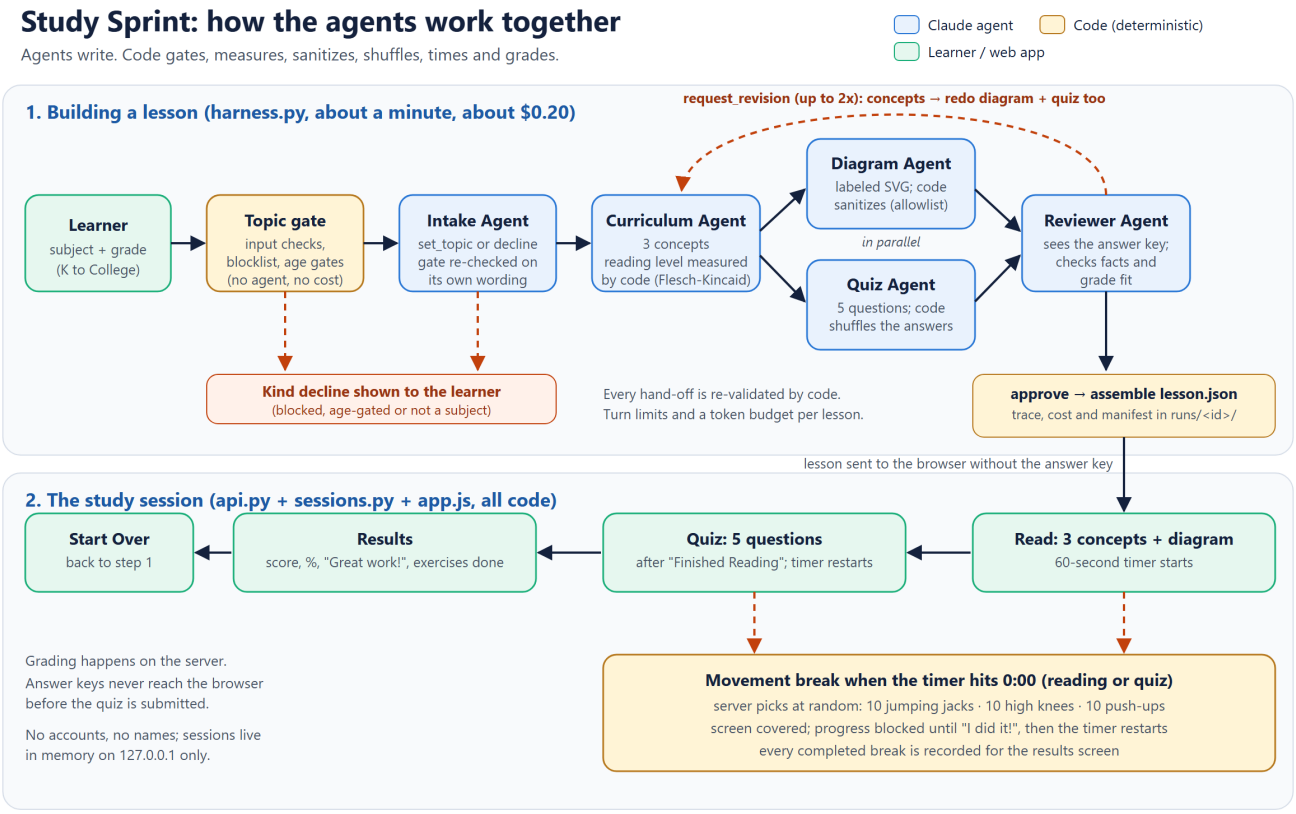
Subjects that aren't right for the learner's grade are declined kindly before any agent runs (here, an age-gated topic for Grade 2).

Works on phones



The layout stacks the concepts on narrow screens.

How the agents build a lesson



Agent	Job	What code checks before accepting its work
Intake	Turns the subject into a specific topic for the grade, or declines it kindly.	The topic gate runs again on the agent's topic, so a blocked subject can't be reworded through.
Curriculum	Writes the three concepts, each with an example.	Exactly 3; word counts per grade band; reading level (Flesch-Kincaid, computed by code); sentence length; safety checks.
Diagram	Draws one labeled SVG diagram with alt text.	Element and attribute allowlist (no scripts, links or styles); size limits; at least 3 labels.
Quiz	Writes 5 questions: a correct answer, 3 distractors and an explanation each.	Distinct options; every concept tested; reading level; safety. Code shuffles answer positions.
Reviewer	Checks facts, the answer key and grade fit.	May send one part back, up to twice; if it still isn't right, the build fails instead of publishing. Revised concepts also redo the diagram and quiz.

The **harness** (plain Python) runs the stages in order, runs Diagram and Quiz in parallel, re-checks every hand-off, enforces turn limits and a token budget, and records a trace, cost and manifest for every build in `runs/`. Exercise choice, timers and grading are code, never agents.

Guardrails

Protects against	How
Unsafe or adult subjects	Hard blocklist and age-gated topics (code), then the Intake Agent's judgment.
Prompt injection	Subject length and character allowlist; injection phrases rejected; the subject is passed to agents as quoted data.
Unsafe lesson text	Every string checked for links, emails, phone numbers, requests for personal information, unkind words and HTML.
Malicious diagrams	SVG allowlist and re-serialization; shown as an image, where browsers never run scripts; strict Content-Security-Policy.
Too hard / too easy	Reading level measured by code against the grade's range.
Wrong answers	Independent Reviewer Agent; grounding eval; answer positions shuffled by code.
Cheating / tampering	Answer keys stay on the server; phases only move forward; a pending break blocks progress.
Runaway cost	Turn limits, a per-lesson token budget, at most 3 builds at once.
The computer it runs on	CLAUDE.md rules: no `os` module, every path confined to the project, no deletions, caches inside the project, server on 127.0.0.1 only.

Evals

Unit, API and browser tests (offline, free) cover every rule, the session flow and the full browser experience with the clock fast-forwarded. **Lesson scorers** re-check any saved build (structure, reading level, safety, SVG, quiz grounding, diagram relevance, review). The **red-team** suite attacks the guardrails and checks that benign look-alikes still pass. The **live matrix** builds lessons from Kindergarten to College with the real agents.

Run it

`.\run.ps1 serve --demo` sample lessons, no API key
`.\run.ps1 serve` live agents (needs ANTHROPIC_API_KEY)
`.\run.ps1 test | .\run.ps1 eval | .\run.ps1 guide`
Then open http://127.0.0.1:8000.